

Tigran Melkonyan

+374(94)77-30-37 | tigranmel2004@gmail.com

EDUCATION

Main

American University of Armenia [link] , [transcript] <i>Bachelor of Science in Computer Science</i>	Yerevan, Armenia Aug. 2023 – Present
Yerevan State University [link] , [transcript] <i>Informatics and Applied Mathematics</i>	Yerevan, Armenia Sep. 2022 – Jun. 2023
Armavir High School № 1 named after Ruben Yeghoyan [link]	Armavir, Armenia Sep. 2019 – May. 2022
Araks Secondary School named after Zorik Avetisyan [link]	Armavir, Armenia Sep. 2010 – May. 2019

Other courses

Mathematical Analysis III, Moscow State University (Online) <i>Supervisor - Stanislav Shaposhnikov</i>	Sep. 2023 – Dec 2023
Analytic Geometry, Yerevan State University	Sep. 2022 – Dec. 2022

OLYMPIADS

- The ICPC International Collegiate Programming Contest (Northern Eurasia Finals) Dec, 2024, [\[PDF\]](#), [\[PDF\]](#)
- The ICPC International Collegiate Programming Contest (Armenia, Regional Contest), Nov, 2024 [\[PDF\]](#), [\[PDF\]](#)
- Mathematics Olympiad held at Yerevan State University, 2022 [\[PDF\]](#)

PROFESSIONAL EXPERIENCE

Center for Scientific Innovation and Education [link] <i>Research Intern</i> <i>Junior Researcher</i>	Yerevan, Armenia Mar. 2025 – Aug. 2025 Aug. 2025 – Present
--	--

- Contributing to experimental research on communication delays, and control-theoretic methods for synchronized drone operations.

RESEARCH PROJECTS

- Reaction-Diffusion PDE Modeling of Epidemiological Systems**
—Formulated and numerically solved a reaction-diffusion SIR system. Discretized using finite-difference methods and Laplacian approximations, visualized virus spread, and compared numerical and analytical solutions.

PUBLICATIONS AND MANUSCRIPTS

- T. Melkonyan, A. Petrosyan, D. Hakobyan, S. Kumar.
Comparative Study of Automated Lung Nodule Detection in Chest X-Ray Images Using Pretrained Deep Learning Models,.
Machine Learning for Computational Science and Engineering, Published 2026.
- Conducted a full experimental study combining CNN-based classification and Faster R-CNN detection for lung nodule analysis on ChestX-ray14 and JSRT datasets.